From Coalition Logic to Rho Calculus: Toward a Curry–Howard Correspondence for Consensus Protocol Synthesis

Research Note

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April 2026

Abstract

We examine the programme of Jamie Gabbay and collaborators on *semitopologies* and *Coalition Logic*, a three-valued modal fixed-point logic for the declarative specification of distributed consensus protocols. We observe that Coalition Logic currently lacks a formal proof system (sequent calculus or natural deduction), and that the published correctness arguments for Declarative Paxos proceed by semantic entailment from axioms rather than by syntactic derivation. We propose that a proof system can be systematically extracted from the model-theoretic semantics using established methods (labelled sequent calculi, coalgebraic logic, multilateral sequents for three-valued reasoning). We then outline a Curry–Howard term assignment in which proofs of Coalition Logic formulae are decorated with terms of a rho calculus variant equipped with *join patterns*, so that the join $\mathsf{for}(y_1 \leftarrow x_1 \ \& \ \cdots \ \& \ y_n \leftarrow x_n)\, P$ witnesses the coalition modality $\boxminus\varphi$. If carried through, this correspondence would yield a pipeline from declarative consensus specifications to correct-by-construction rho calculus implementations.

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1 Introduction

Distributed consensus is the foundational problem of blockchain and decentralised systems: how can heterogeneous, potentially faulty participants agree on a shared state without a central authority? Existing approaches to specifying and verifying consensus protocols—most prominently TLA+ [Lam02]—describe state transitions in a sequential, step-by-step manner. This operational style is powerful for model checking but obscures the *logical essence* of the protocol.

Gabbay and collaborators [Gab23, Gab24, GL23, GZ25, Gab25] have developed a strikingly different approach grounded in topology. Their programme proceeds in three layers:

(i) **Semitopologies** [Gab23, Gab24]: a generalisation of point-set topology in which the intersection of two open sets need not be open. Points are participants; open sets are *actionable coalitions* (quorums).

(ii) **Semiframes** [GL23]: an algebraic dual to semitopologies, analogous to the classical frame/topology duality, but with a compatibility relation $*$ replacing set intersection.

(iii) **Coalition Logic** [GZ25, Gab25]: a three-valued modal fixed-point logic whose models are semitopologies, in which a distributed algorithm is presented as a logical theory (signature plus axioms) and correctness properties are derived from those axioms.

The present note addresses the following question:

Question. *Can one build, on the theoretical basis of semitopologies and Coalition Logic, a logic-based language for specifying consensus protocols from which it is possible to synthesise a high-performance implementation?*

We argue that the answer is *yes in principle*, contingent on filling a specific gap in the existing theory: Coalition Logic currently has no formal proof system. We outline a strategy for:

(a) extracting a proof system from the model-theoretic semantics;

(b) establishing a Curry–Howard correspondence between proofs in that system and terms of the rho calculus with join patterns;

(c) using cut elimination / proof normalisation as the computational engine that transforms declarative specifications into executable protocol implementations.

2 Gabbay's Programme: Semitopologies, Semiframes, and Coalition Logic

2.1 Semitopologies

Definition 2.1 (Semitopology [Gab23])**.** A *semitopology* is a pair (P, Open) where P is a set of *participants* and $\mathsf{Open} \subseteq \mathcal{P}(P)$ is a collection of *open sets* (actionable coalitions) satisfying:

(1) $\emptyset \in \mathsf{Open}$ and $P \in \mathsf{Open}$;

(2) Open is closed under arbitrary unions.

Unlike a topology, closure under finite intersection is *not* required.

The key intuition is that just because two coalitions each have sufficient resources to act does not mean their overlap does. This single relaxation opens up a rich theory that departs rapidly from classical topology.

Central to the analysis of consensus are *antiseparation properties*. A semitopology is *n-twined* if every n nonempty open sets have nonempty intersection—this generalises the quorum intersection property used in the correctness proofs for Paxos and its variants. Gabbay introduces a classification of points (regular, weakly regular, quasiregular) and proves that every semitopology partitions into *topen sets* (transitive open sets that are guaranteed to reach agreement) plus points of other kinds.

2.2 Semiframes

Semiframes [GL23] are the algebraic counterpart of semitopologies, standing in the same relation to them as frames stand to topological spaces. A semiframe is a complete join-semilattice equipped with a *compatibility relation* $*$ that generalises set intersection. Gabbay proves a categorical duality between sober semiframes and spatial semitopologies.

The surprising finding is that the canonical algebraic structure requires $*$ rather than a meet operation—the algebra of semitopologies is not a lattice in the expected sense. This compatibility relation turns out to be precisely what is needed to express well-behavedness properties (such as n-twinedness) algebraically.

2.3 Coalition Logic

Definition 2.2 (Coalition Logic, informal [GZ25])**.** Coalition Logic is a three-valued modal fixed-point logic. Its models are semitopologies (P, Open) equipped with an interpretation of predicate symbols as functions $P \to \mathbf{3}$, where $\mathbf{3} = \{\mathbf{f} < \mathbf{b} < \mathbf{t}\}$ is Kleene's strong three-valued lattice. The central modality is

$$[\![\exists\Box\varphi]\!](p) = \mathbf{t} \quad \text{iff} \quad \exists\, O \in \mathsf{Open}.\; p \in O \;\wedge\; \forall\, q \in O.\; [\![\varphi]\!](q) = \mathbf{t}$$

expressing: "there exists an actionable coalition containing p at which φ holds everywhere." A dual reading of the third truth value $\mathbf{b}$ captures faulty/indeterminate participants.

A distributed algorithm is specified as a *logical theory* $\mathcal{T} = (\Sigma, \mathsf{Ax})$: a signature Σ of predicate symbols together with a set of axioms Ax expressed in Coalition Logic. Correctness properties are then *derived from* the axioms.

Gabbay and Zanolini demonstrate this on Paxos [GZ25], and Gabbay applies the method to voting, Bracha Broadcast, and Crusader Agreement [Gab25]. In ongoing work, the techniques have been used to find errors in a proposed industrial protocol (Heterogeneous Paxos).

3 The Gap: No Proof System

The word "derived" in Gabbay's work means *semantic entailment*: the published correctness proofs for Declarative Paxos are mathematical arguments that the axioms of the theory (Figure 10 of [GZ25]) semantically imply the desired correctness predicates. There is no:

- sequent calculus,
- natural deduction system, or
- Hilbert-style deductive calculus

for Coalition Logic. This is a deliberate scoping choice—Gabbay's programme has focused on semantics first—but it is the critical missing piece for any Curry–Howard-style synthesis strategy.

Without a proof system, there is no notion of *proof object*, and therefore no target for term assignment.

4 Extracting a Proof System from the Semantics

The model-theoretic semantics of Coalition Logic is Kripke-style (broadly construed), and there are well-developed methodologies for systematically constructing sound, complete, and cut-free proof systems from such semantics. We survey the principal options.

4.1 Labelled Sequent Calculi

Following Negri and von Plato [NvP01], one internalises the semantic parameters into the sequent syntax. Sequents take the form

$$\mathcal{R},\ \Gamma\ \vdash\ \Delta$$

where $\mathcal{R}$ is a set of *relational atoms* encoding semitopological structure—atoms such as $p \in O$, $O \in \mathsf{Open}$, $O \subseteq O'$—and Γ, Δ are multisets of *labelled formulae* $p : \varphi$.

The semantic clause for $\exists\Box\varphi$ translates into introduction rules:

$$\frac{\mathcal{R},\ O \in \mathsf{Open},\ p \in O,\ \{q : \varphi \mid q \in O\}\ \vdash\ \Delta}{\mathcal{R}\ \vdash\ p : \exists\Box\varphi,\ \Delta}\ \exists\Box\mathrm{R}$$

with an eigenvariable condition on O. Negri's methodology guarantees that any semantic condition expressible as a *geometric* ($\forall\exists$-Horn) formula over the frame can be turned into a structural rule preserving cut elimination. Many of Gabbay's well-behavedness conditions (n-twinedness, regularity, antiseparation) are geometric or near-geometric, making this approach directly applicable.

4.2 The Coalgebraic Approach

The semiframe duality of [GL23] provides an algebraic semantics that fits naturally into the coalgebraic logic programme of Pattinson and Schröder [PS10]. In this framework:

- The modality $\exists\Box$ is characterised by a functor on a suitable category.
- The *one-step semantics*—how $\exists\Box\varphi$'s truth value depends on φ's truth values across the coalgebra—determines the modal rules of a sequent calculus.
- A generic cut-elimination theorem applies provided the one-step rules satisfy an *absorption* condition.

The compatibility relation $*$ on semiframes would appear as a structural connective mediating interaction between coalition modalities.

4.3 Handling Three-Valuedness

The three-valued base $\mathbf{3} = \{\mathbf{f}, \mathbf{b}, \mathbf{t}\}$ requires adaptation. Two approaches are available:

Multilateral sequent calculi. Following Rumfitt and Incurvati–Schlöder, sequents have three zones—one per truth value. A sequent asserts the inconsistency of the combined assignment. This directly encodes the three-valued semantics and has natural proof-theoretic properties.

Bilattice-based calculi. Avron's programme for bilattice logics provides cut-free sequent calculi for the class of logics that includes Kleene's strong three-valued logic. Since $\mathbf{3}$ is a bilattice (in fact, a product bilattice $\mathbf{2} \times \mathbf{2}$ modulo a quotient), Avron's methods apply directly.

4.4 Fixed Points: Cyclic Proofs

Coalition Logic includes fixed-point operators, placing it in the territory of the modal μ-calculus. For proof-theoretic purposes, *cyclic proof systems* [Bro06, DMV23] are the right tool: proofs are finite directed graphs with a global soundness condition (every infinite path through the unfolded proof passes through a regeneration point). These admit cut elimination and, crucially, correspond naturally to recursive/corecursive programs under Curry–Howard—exactly the reactive, looping processes that consensus protocols require.

5 A Curry–Howard Correspondence for Rho with Joins

5.1 The Rho Calculus with Joins

We work with a variant of the rho calculus (Reflective Higher-Order calculus) [Mer05] extended with *join patterns*:

$$P, Q \ ::= \ \mathbf{0} \ \mid \ \mathsf{for}(y_1 \leftarrow x_1 \ \& \ \cdots \ \& \ y_n \leftarrow x_n)\, P \ \mid \ x!\langle M \rangle \ \mid \ P \mid Q \ \mid \ \mathsf{contract}\ \mathsf{for}(\vec{y} \leftarrow \vec{x})\, P \ \mid \ *x$$

where $*x$ is the rho calculus dereferencing (quoting/unquoting) operator. The join pattern $\mathsf{for}(y_1 \leftarrow x_1 \ \& \ \cdots \ \& \ y_n \leftarrow x_n)\, P$ synchronises on n channels simultaneously: the continuation P fires only when all channels $x_1, \ldots, x_n$ have a message available.

5.2 The Proposed Correspondence

The structural match between Coalition Logic and rho-with-joins is as follows.

Coalition Logic	**Rho Calculus with Joins**
Participant p	Channel name x_p
Open set $O = \{p_1, \ldots, p_n\}$	Channel set $\{x_{p_1}, \ldots, x_{p_n}\}$
$\exists\Box\varphi$ at p	$\mathsf{for}(y_1 \leftarrow x_{p_1} \ \& \ \cdots \ \& \ y_n \leftarrow x_{p_n})\, P_\varphi$
φ with value $\mathbf{t}$	Running term (reduces normally)
φ with value $\mathbf{b}$	Blocked/stuck term (faulty participant)
φ with value $\mathbf{f}$	Dead/impossible term
n-twinedness	Channel-sharing constraint between joins
Fixed-point unfolding	$\mathsf{contract}$ (replicated input)

The proposed term assignment decorates the proof rules of §4 as follows.

Coalition introduction ($\exists\Box$R). The right-introduction of the coalition modality introduces a join pattern:

$$\frac{\{q : \varphi \ \triangleright \ P_q\}_{q \in O}}{p : \exists\Box\varphi \ \triangleright \ \mathsf{for}(y_1 \leftarrow x_{p_1} \ \& \ \cdots \ \& \ y_n \leftarrow x_{p_n})\, P} \ \exists\Box\mathrm{R}$$

where $O = \{p_1, \ldots, p_n\}$ is the witnessing open set and P is assembled from the individual witnesses P_q.

Coalition elimination ($\exists\Box$L). The left-elimination (using a coalition hypothesis) introduces a send:

$$\frac{p : \varphi \ \triangleright \ M}{p : \exists\Box\varphi \ \triangleright \ x_p!\langle M\rangle}\ \exists\Box\text{L}$$

A participant who knows $\exists\Box\varphi$ can contribute their local evidence by sending on their channel.

Cut = Communication. Cut elimination in labelled modal calculi corresponds to message passing. A cut on $p : \exists\Box\varphi$ between a right-introduction (a join waiting for coalition messages) and a left-elimination (a send contributing to the coalition) reduces to the actual communication step of the protocol. The cut-free proof is the final agreed-upon state.

Fixed points = Replicated processes. A cyclic proof, in which a branch returns to an earlier sequent, is decorated with a contract pattern—a replicated input that re-enables the join after each round of consensus.

5.3 The Three-Valued Dimension

The third truth value **b** has a natural process-algebraic interpretation:

- A term in the **t**-zone is a *correct* process—it reduces and produces the expected output.
- A term in the **b**-zone is a *faulty* process—it may behave arbitrarily (Byzantine) or may simply be stuck.
- A term in the **f**-zone is an *impossible* process—it can be shown to never reduce.

The fundamental logical property of n-twined semitopologies—that any two quorums share a correct (**t**) participant—translates into the process-algebraic guarantee that any two join patterns share a channel carrying a well-typed message. This is the computational content of the quorum intersection lemma.

6 The Synthesis Pipeline

If the programme of §4–§5 is carried through, the resulting synthesis pipeline has three stages.

Stage 1: Specification. A consensus protocol is specified as a Coalition Logic theory $\mathcal{T} = (\Sigma, \mathsf{Ax})$ in the style of Gabbay–Zanolini. Correctness properties are stated as formulae to be proved from Ax.

Stage 2: Proof and extraction. A proof of the correctness properties in the labelled cyclic sequent calculus of §4 is constructed (possibly with automated assistance). The Curry–Howard correspondence of §5 extracts a rho calculus term from the proof. This term is *correct by construction*: it implements the protocol and satisfies the stated correctness properties by virtue of its provenance as a proof decoration.

Stage 3: Optimisation. Curry–Howard extraction typically produces naive, unoptimised code. For consensus protocols, the critical performance metrics are message complexity and round complexity. Optimisation proceeds by:

- applying bisimulation-preserving transformations to the extracted rho term, reducing communication overhead while maintaining correctness;
- using the phlogiston (phlo) metering of the rholang runtime as an objective function for optimisation;
- compiling the optimised rho term to the rholang bytecode interpreter for execution on a F1R3FLY shard.

7 Connections and Context

7.1 Relation to Spatial Logics for Process Calculi

Coalition Logic's modality $\exists\Box$, quantifying over open sets in a semitopology, has a structural resemblance to spatial modalities in logics for the ambient calculus and the π-calculus [CC01, CC04]. In spatial logics, a formula $\varphi \mid \psi$ asserts that the current process can be decomposed into parallel components satisfying φ and ψ respectively. Coalition Logic's open-set quantification plays a similar decomposition role, but over *participants* rather than *process structure*. A systematic comparison may yield insights in both directions.

7.2 Relation to MeTTaIL

MeTTaIL (rholang 1.4) provides a framework for defining languages as collections of types (syntactic categories), equations on types, and rewrites on types. Coalition Logic axioms could potentially be expressed as MeTTaIL rewrite rules, with the semitopological quorum structure encoded in the type/grammar layer. The stochastic and quantum simulation capabilities of the `gillespie` crate could be used to explore the probabilistic aspects of consensus protocols.

7.3 Gabbay's Proof-Theoretic Background

It is worth noting that Gabbay has extensive proof-theoretic expertise. His earlier work includes a sequent calculus for nominal logic, the Curry–Howard correspondence for Contextual Modal Type Theory (with Nanevski), and the nominal algebra programme. The absence of a proof system for Coalition Logic appears to be a deliberate scoping decision rather than an oversight, and Gabbay would be a natural collaborator for the programme outlined here.

8 Open Problems and Future Work

P1. Proof system for Coalition Logic. Construct a labelled cyclic sequent calculus for Coalition Logic, incorporating multilateral sequents for three-valuedness and cyclic proofs for fixed points. Prove cut elimination.

P2. Geometric characterisation of semitopological conditions. Determine which of Gabbay's well-behavedness conditions (n-twinedness, regularity, weak regularity, quasiregularity) are expressible as geometric implications, and hence convertible to structural rules preserving cut elimination via Negri's method.

P3. Coalgebraic analysis of the coalition modality. Identify the functor on semiframes (or a suitable category) that characterises the $\exists\Box$ modality, and apply the Pattinson–Schröder generic cut-elimination framework.

P4. Curry–Howard term assignment. Define the term assignment of §5.2 precisely and prove that it satisfies subject reduction (typing is preserved under cut elimination / β-reduction) and progress (well-typed open terms reduce or are values).

P5. Adequacy of extracted terms. Show that rho terms extracted from proofs of Coalition Logic correctness properties are *adequate* implementations of the specified protocol, in the sense that their observable behaviour (under a suitable notion of observation) matches the semantic content of the proved properties.

P6. Optimisation preserving bisimulation. Develop bisimulation-preserving optimisation strategies for extracted rho terms, with phlogiston cost as the objective function.

P7. **Mechanisation.** Formalise the proof system and term assignment in a proof assistant (Lean 4, Coq, or Agda), enabling machine-checked synthesis.

9 Conclusion

Gabbay's programme on semitopologies and Coalition Logic provides two-thirds of the infrastructure needed for logic-based consensus protocol synthesis: a rich mathematical semantics (semitopologies and semiframes) and a genuine declarative specification language (Coalition Logic) that has been validated on nontrivial protocols and has found real errors in proposed industrial designs.

The missing third—a formal proof system admitting a Curry–Howard correspondence with the rho calculus—is not yet published but is extractable from the existing semantics using well-established methods. The structural match between the coalition modality $\exists\Box$ and the rho calculus join pattern $\mathsf{for}(y_1 \leftarrow x_1 \ \& \ \cdots \ \& \ y_n \leftarrow x_n)\, P$ is surprisingly tight, and the interpretation of cut elimination as message passing gives a natural computational dynamics.

The programme outlined here would, if successful, close the loop from declarative consensus specifications to correct-by-construction implementations, with formal guarantees rooted in proof theory.

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